University of kasdi Merbah Ouargla
Faculty of New Technologies of Information and Communication
Department of computer science and Information Technology

Module: Electronics, system components

Corrected Series of TD n° 6

Exercise 1:

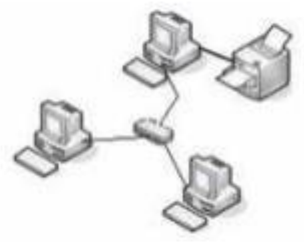
1) The advantages of computer networks are:
− Share hardware and software resources between network members (files, applications, printer, scanner, modem, etc.)
− Communication between people (e-mail, live chat, etc.)
− Save time
− Reduce equipment cost
2) The different networks classified according to geographic coverage (distance) are:
− Personal networks (PAN: Personal Area Network)
− Local networks (LAN: Local Area Network)
− Metropolitan Area Networks (MAN: Metropolitan Area Network)
− Wide area networks (WAN: Wide Area Network)
The different networks classified according to architecture are:
− Point to Point (Peer to Peer)
− Clients/Servers
3) The hardware components that can be found in a network:
− Physical media (cables, physical links, transmission lines, medium, transmission channel, etc.)
− Interconnection equipment (switches, routers, etc.)
− Terminal equipment (computers, stations, servers, peripherals, host machines, etc.).
4) The type of network (LAN, MAN, WAN or PAN) to use for each of the following cases:

Description	Network
A network that connects computers in a building.	LAN
A network that connects staff devices (laptop, smartphone, etc.).	PAN
A network that connects a company's subsidiaries internationally.	WAN
A network that connects the subsidiaries of a company located in a city.	MAN

Exercise 2:

1)
a) Router
b) Concentrator (HUB)
c) Switch (SWITCH)

2)
a) So that all three computers can print to the same printer:
− Connected together by a switch (create a network using switch)
− Test the connectivity using ping @ip
− Share the printer on the network
b) Solution diagram:



Exercise 3: Choose the right answer:

1. Network devices link together using a variety of connections like:
   Copper cabling
2. Benefits from networking include:
   Centralized administration
3. A group of interconnected computers under one administrative control group that governs the security and access control policies that are in force on the network.
   LAN
4. A group of wireless devices that connect to access points within a specified area. Access points are typically connected to the network using copper cabling.
   WLAN
5. Network that connects devices, such as mice, keyboards, printers, smartphones, and tablets within the range of an individual person. They are most often connected with Bluetooth technology.
   PAN
6. Network that spans across a large campus or a city. Consisting of various buildings interconnected through wireless or fiber optic backbones.
   MAN
7. Connections of multiple smaller networks such as LANs that are in geographically separated locations. The most common example of a WAN is the Internet.
   WAN
8. Devices which are connected directly to each other without any additional controlling devices between them. Each device has equivalent capabilities and responsibilities.
   Peer-to-peer networks
9. In a client/server model, the client requests information or services from the server. The server provides the requested information or service to the client.
   Client/server networks
10. An IP address is a unique number that is used to identify a network device and is represented as a ------------- bits binary number, divided into -------- octets (groups of eight bits):
    32 , 4
11. Convert 11111101 to decimal
    253
12. Convert 24 to binay
    00011000
13. To determine the IP address for a certain computer using CMD tool , you type the command:
    ipconfig

Good luck
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